MIL-L-19661A
22 MAY 1962
SUPERSEDING DATA
(SEE 6.2)

MILITARY SPECIFICATION

LIGHT, MARKER, PORTABLE, EMERGENCY, AIRFIELD, BATTERY OPERATED

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 *Scope.* This specification covers portable marker lights for general emergency use and on ground obstructions, airfield runways and taxiways, and landing areas for rotary wing aircraft.

1.2 Classification. Marker lights shall be of the following types, as specified (see 6.3):

Type I	— Dry battery operated, precision optical system.
Type II	— Mercury battery operated, precision optical system.
Type IIIA	— Dry battery operated, nonprecision optical system, two-position switch.
Type IIIB	— Dry battery operated, nonprecision optical system, three-position switch, flash capability.
Type IV	— Dry battery operated, precision optical system, flashing red light.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-P-636	— Primer Coating, Synthetic, Wood and Ferrous Metal.
TT-P-664	— Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-B-601	— Boxes, Wood, Cleated-Plywood.
PPP-B-621	— Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	— Box, Fiberboard.
PPP-B-665	— Boxes; Paperboard, Metal Stayed (Including Stay Material).

FSC 6230

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

PPP-B-676 — Boxes, Set-up, Paperboard.

MILITARY

MIL-B-18 — Batteries, Dry.

MIL-P-116 — Preservation, Methods of.

MIL-S-5002 — Surface Treatments (Except Priming and Painting) for Metal and Metal Parts in Aircraft.

MIL-E-5272 — Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

MIL-E-5556 — Enamel, Camouflage, Quick Drying.

MIL-C-7989 — Covers, Light-Transmitting, for Aeronautical Lights, General Specification for.

MIL-C-8514 — Coating Compound, Metal Pretreatment, Resin-Acid.

MIL-P-8585 — Primer Coating, Zinc Chromate, Low-Moisture - Sensitivity.

MIL-T-9107 — Test Reports, Preparation of.

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL-C-25050 — Colors, Aeronautical Lights and Lighting Equipment, General Requirements for.

MIL-D-70327 — Drawings, Engineering and Associated Lists.

STANDARDS

FEDERAL

FED-STD- No. 595 — Colors

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking for U. S. Military Property.

MS33586 — Metals, Definition of Dissimilar.

DRAWINGS

CORPS OF ENGINEERS

13207E8810 — Light, Marker, Ground Obstruction, Emergency, Airfield Runway, Portable, Battery-Operated (Type IIIA).

13207E8820 — Light, Marker, Ground Obstruction, Emergency, Airfield Runway, Portable, Battery-Operated (Type IIIB).

13207E8840 — Bag, Carrying, Light Marker, Assembly.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 **Preproduction.** This specification makes provision for preproduction inspection (see 4.3).

3.2 **Materials.** Materials shall conform to applicable specifications as specified herein and on the applicable drawings. Materials which are not covered by applicable specifications or drawings or which are not specified herein shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended. The material shall be free from all defects and imperfections that may affect the quality and reliability of the finished product.

3.2.1 *Protective treatment.* Materials used in the construction of the lights that are subject to corrosion in oxygen, salt air, or other atmospheric conditions likely to occur during service usage, shall be protected against such corrosion in a manner that will in no way prevent conformance to the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.2.2 *Metals.* All metals used in the construction of the light shall be of a corrosion-resisting type or shall be suitably protected to resist corrosion. The use of dissimilar metals shall be avoided whenever practicable or used in accordance with Standard MS 33586.

3.3 **Commercial parts.** Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

(a) They can be replaced by the military standard parts without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.3.1 *Standard parts.* With the exception of 3.3, military standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 **Design and construction.** The lights shall be designed and constructed in accordance with table I.

3.4.1 *Optional designs.* Optional designs and details may be acceptable provided the interchangeability and performance requirements are met, and if approved by the procuring activity.

3.4.2 *Construction.* The light shall be so constructed that parts will not work loose in service, and repairs and adjustments can be easily made by personnel of operating units without the use of special tools. It shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5 **Components.** The major components of the various types of marker lights shall be as specified in table I and the following subparagraphs.

3.5.1 *Case.* The case for all types of lights shall be constructed of not less than No. 22 gage sheet steel or aluminum. It shall be provided with space and suitable contacts for the specified battery and a means of support to prevent tilting or overturning in severe winds or propeller or jet blast. The case shall be watertight. Types I and II light cases shall be provided with a fixed rack for color filters. Type III light case shall provide space for storage of a spare lamp.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

TABLE I. *Major components and characteristics of marker lights*

Components	Type I	Type II	Type IIIA	Type IIIB	Type IV
Batteries	Dry battery—BA-200/U 4 batteries	Mercury battery BA-1030/V 4 batt.	Dry battery, 6 volts, 1 battery		Dry battery, 1 or more
Flasher	Built-in unit (max)—130 flashes/minute (min)— 60 flashes/minute		None	Built-in unit	Built-in unit 60 ±10 flashes/minute
Switch	3-position. Labeled ON, OFF, and FLASH		2-position. Labeled ON-OFF	3-position. Labeled ON-OFF-FLASH	2-position. Labeled ON and OFF
Lamp	2 GE PR-12, 1 for spare	2 GE PR-13, 1 for spare	1 (trade number) #502		Optional
Lens	Clear Fresnel lens, class D of MIL-C-7989		Nondiffusing plastic dome, class D of MIL-C-7989		Red Fresnel lens class D of MIL-C-7989 and aviation red of MIL-C-25050[1]
Color filters	4 aviation colors-type I, grade B, of MIL-C-25050		4 aviation colors and 1 directional, type I, grade B of MIL-C-25050		None
Hood	To restrict light: (horiz.) (15 to 30 degrees) (vert.) (15 to 30 degrees)		To restrict light 20 degrees solid angle in the horizontal plane		None
Socket	Single contact miniature flanged		Miniature screw		Single contact, prefocus
Drawing lists	...	...	13207E8810 13207E8840	13207E8820 13207E8840	...
Weight	4½ pounds	Hold to minimum	2¾ pounds	2¾ pounds	(+ batteries) 18 pounds
(Size) dimensions	9 by 7½ by 6 in. approx. 420 cu. in.	Hold to minimum	With ballast 9⅛ by 9½ by 11½ inches		1,000 cubic inches
Life (min)	75 hours flashing	50 hours flashing	75 hours	75 hours	250 hours

[1] The transmittance of the lens need not be specified as the intensity of the light with colored lens specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

3.5.2 *Batteries.* Batteries specified in table I shall conform to Specification MIL-B-18.

3.5.2.1 The battery ordering data shall be plainly marked inside the battery compartment.

3.5.2.2 Unless otherwise specified, batteries shall not be furnished with the light (see 6.3).

3.5.3 *Life.* The active life of the light shall be determined as specified in 4.5.4.

3.5.4 *Rain resistance.* When tested as specified in 4.5.6, there shall be no noticeable variation in flashing frequency or light output.

3.5.5 *Humidity.* The light shall operate satisfactorily and shall show no evidence of excessive corrosion when tested as specified in 4.5.7.

3.5.6 *Optical system.*

3.5.6.1 *Precision optical system.* This system shall consist of a lamp, single contact prefocus lamp socket, a lens, and supporting means. It shall be so designed that when the lamp is installed and operated at rated current, the candlepower shall be not less than the values specified in figures 1, 2, and 3, for the specified vertical angles throughout 360 degrees in azimuth, except in sections affected by the bail, where a shadow not exceeding 5 degrees in width shall be permitted. The intensity throughout the main beam shall be as uniform as feasible with best lens design and practice. The lens shall be cylindrical, plastic, Fresnel lens secured in raintight relationship to the light case, and readily detachable.

3.5.6.2 *Nonprecision optical system.* This system shall consist of a clear dome, lamp, lamp socket, and reflector unit, and interchangeable color domes. The light intensity shall be substantially uniform throughout 360 degrees in the horizontal plane and 90 degrees vertically. The candlepower for any angles of view above the horizontal shall be not less than 1.5 candles. The clear domes and the color domes shall be made of class D nondiffusing plastic in accordance with Specification MIL-C-7989. The colors of the dome lenses shall be type I, grade B, or better, in accordance with table titled "Transmission and brightness limits (Illuminant 2,854°K)" of Specification MIL-C-25050.

3.5.7 *Hood.* If required (see table I), a detachable hood shall be furnished by means of which the light emitted shall be restricted to a horizontal angle of not less than 15 degrees nor more than 30 degrees, and a vertical angle of the same dimension. The lower limit of the light cutoff shall be approximately 0 degree vertical angle. The hood shall be designed for installation inside of the lens and when a color filter is used, it shall envelop the color filter. It shall be stowed securely with spare color filters in the case when not in use.

3.5.8 *Switch.* A waterproof, externally operated switch shall be furnished with each light. Switches shall be of the types specified in table I and labeled as specified.

3.5.9 *Socket.* Socket shall be of best commercial grade and shall be firmly mounted in the light case. It shall securely hold the specified lamp within necessary tolerance to insure conformance to the candlepower and beam elevation requirements specified in this specification.

3.5.10 *Lamp.* The quantity and size of the lamp furnished with each light shall conform to table I.

3.5.11 *Flasher.* The flashing frequency shall be as specified in table I when the light is operated with new batteries of the type specified in any ambient temperatures between 0° C and 55° C.

3.5.12 *Filters.* The chromaticity and transmittance of the filters and colored lenses or covers shall be as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

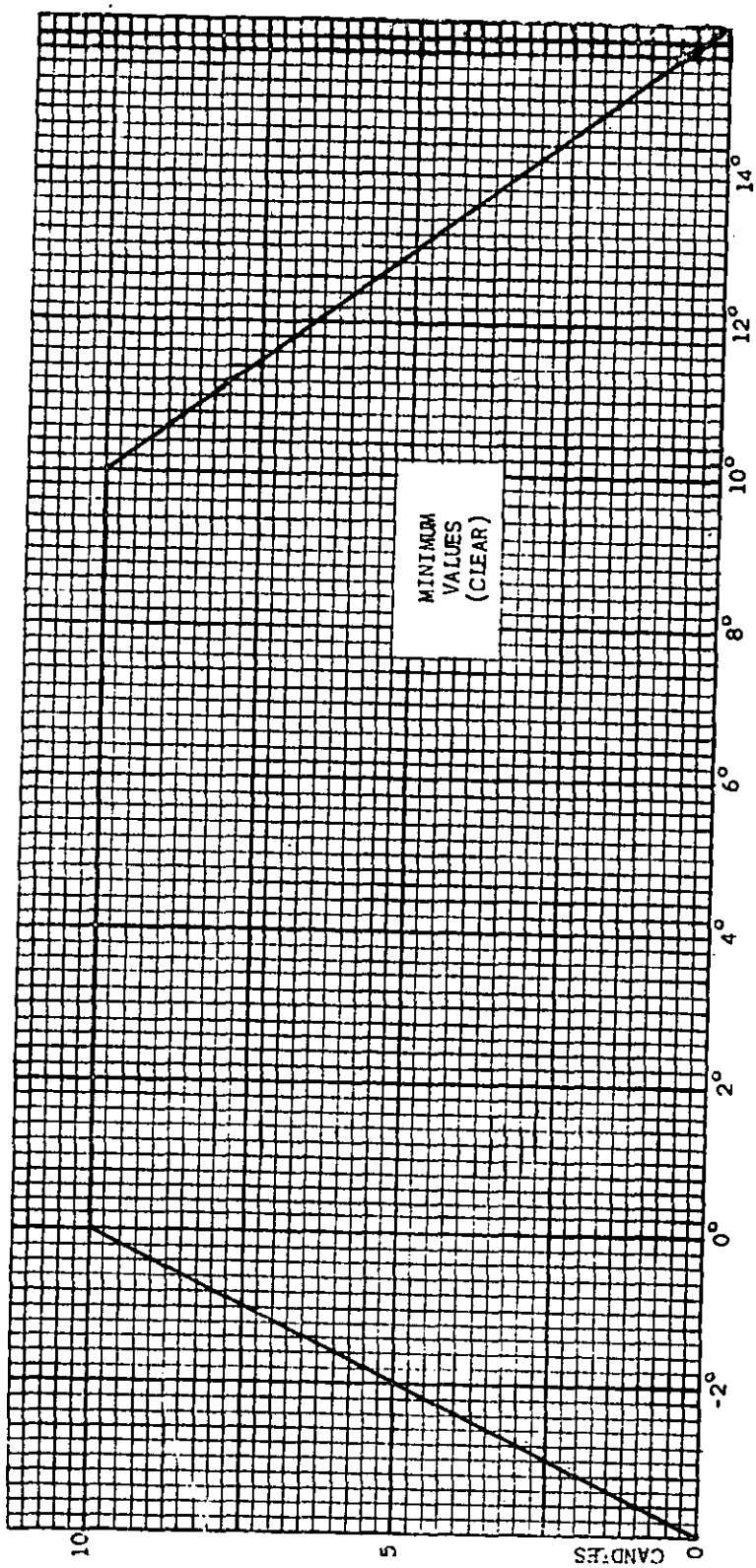


FIGURE 1. *Light distribution, type I light.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

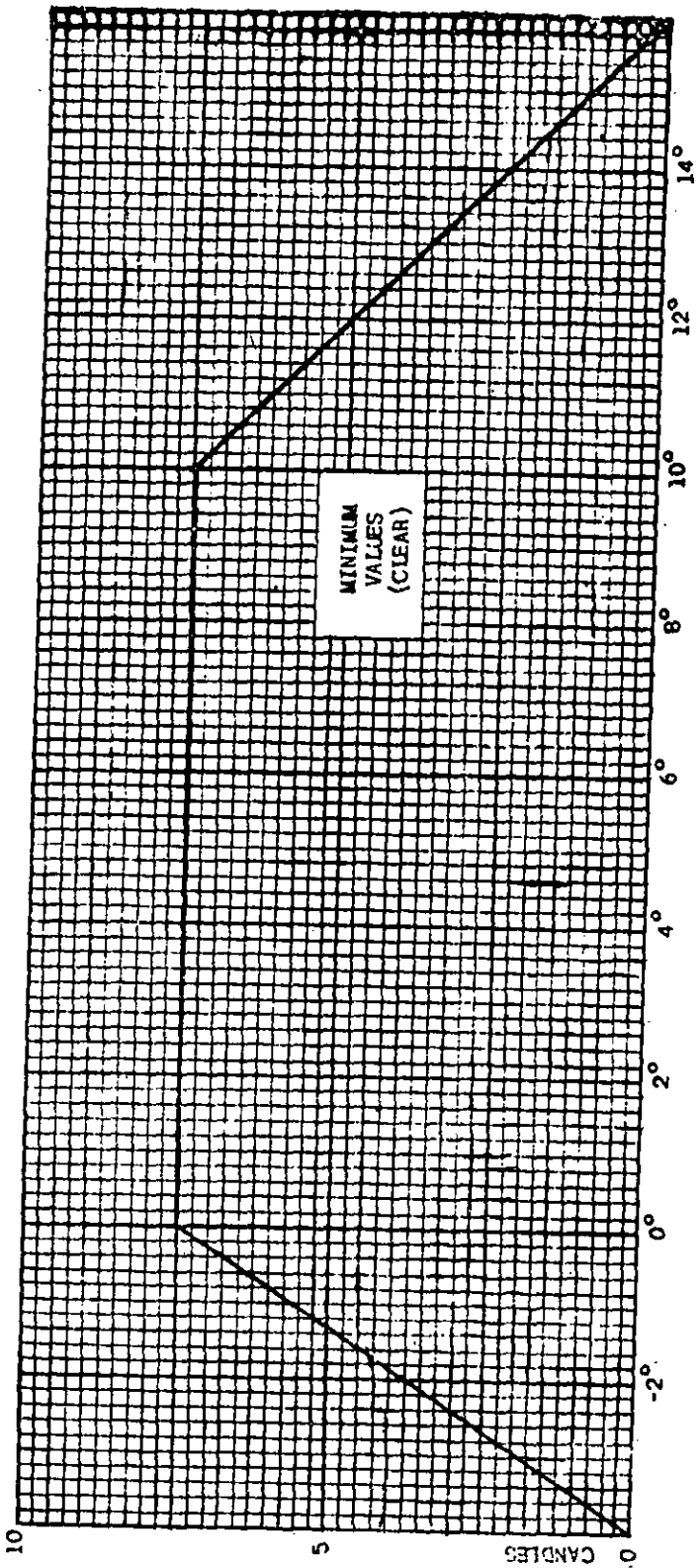


FIGURE 2. *Light distribution, type II light.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

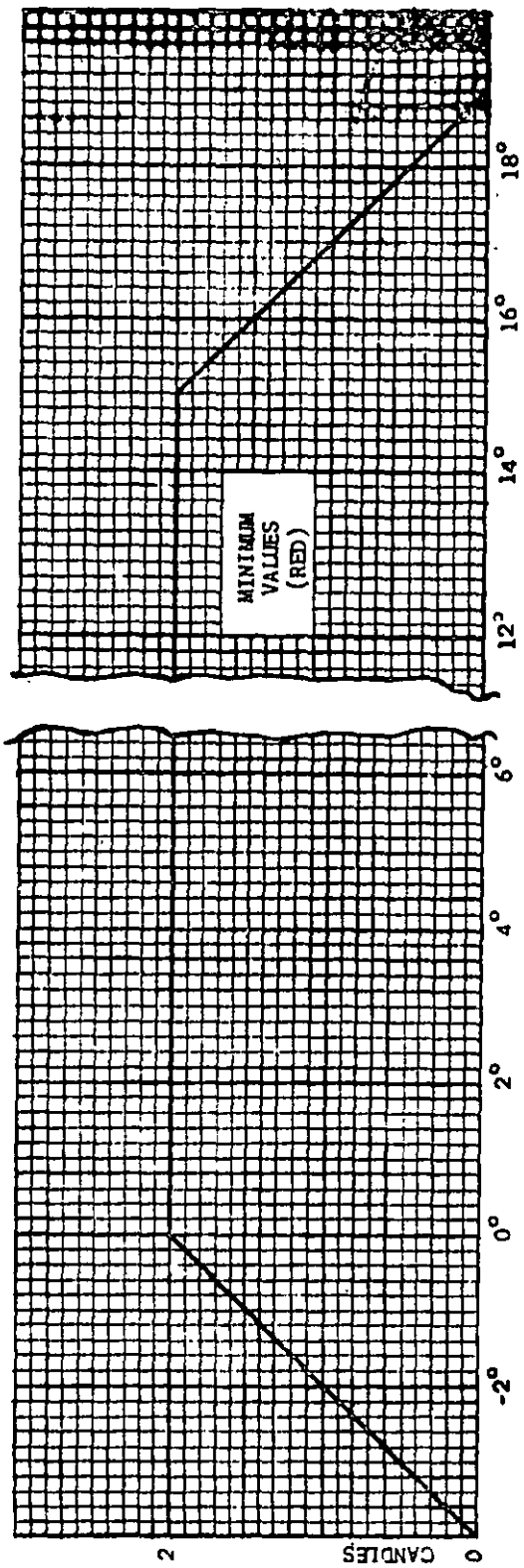


FIGURE 3. *Light distribution, type IV light.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

3.6 **Dimensions.** The total volume of the cylinder or rectangular parallelepiped which can contain the light shall be not greater than 1,000 cubic inches.

3.7 **Interchangeability.** All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.8 **Coating.** The case bottom, top, catches, and hood shall be finished inside and out with a protective system. The cleaning and surface treatment shall be in accordance with Specification MIL-S-5002. One coat of wash primer in accordance with Specification MIL-C-8514 (0.2 to 0.3 mil thick), two coats of primer conforming to Specification MIL-P-8585, TT-P-636, or TT-P-664 (0.6 to 0.8 mil thick), and one coat of orange-yellow enamel (1.0 mil thick), Color No. 33538 of Federal Standard No. 595, conforming to Specification MIL-E-5556, shall be applied.

3.9 **Identification of product.**

3.9.1 The lights shall be marked for identification in accordance with Standard MIL-STD-130.

3.10 **Workmanship.** The light, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to insure freedom from blemishes, defects, burrs and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts and tightness of assembly screws and bolts.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Classification of inspection.** The inspection of marker lights shall be classified as follows:

(a) Preproduction inspection (4.3)

(b) Acceptance inspection (4.4)

4.3 **Preproduction inspection.**

4.3.1 *Preproduction inspection sample.* Unless otherwise specified, as soon as practicable after award of contract or order and prior to submission of any complete lights for final acceptance, the contractor shall furnish a preproduction test sample, consisting of one complete light of each type produced under each contract, for testing to determine conformance to this specification. The approval of the preproduction sample by the activity concerned shall not relieve the contractor of his obligation to supply equipment conforming to this specification. Any changes or deviations from the preproduction sample shall be approved by the contracting officer. The preproduction sample shall be accompanied by the contractor's drawings and shall be tested at a location specified by the procuring activity (see 6.3). The preproduction inspection sample shall be plainly identified by securely attached durable tags marked with the following information:

LIGHT, MARKER, PORTABLE, EMERGENCY, AIRFIELD, BATTERY OPERATED

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

Type (as applicable)

Contract No.

Manufacturer's name

Manufacturer's Part No.

The contractor shall furnish required batteries with the preproduction sample only.

4.3.1.1 *Manufacturer's drawings.* Manufacturer's drawings submitted with the preproduction test sample shall have a cutaway section showing all parts and their normal assembled position and shall specify part numbers of all parts and assemblies. The drawings shall be in accordance with Specification MIL–D–70327.

4.3.2 *Preproduction tests.* The preproduction tests shall consist of all the tests of this specification as specified in 4.5.

4.3.3 *Preproduction test report.* After the contractor completes the preproduction tests, he shall prepare a preproduction report in accordance with Specification MIL–T–9107 and furnish two complete copies to the procuring activity (see 6.3). The test report shall include the following:

(a) Report of all tests

(b) Outline and description of tests and test conditions

4.4 **Acceptance inspection.** Acceptance inspection shall consist of individual tests and sampling tests.

4.4.1 *Individual tests.* Each marker light shall be subjected to the test specified in 4.5.1.

4.4.2 *Sampling tests.* One marker light shall be selected at random from each lot of 50 or fraction thereof of similar lights on the order and subjected to the tests specified in 4.5.2 and 4.5.5.

4.4.2.1 *Lot.* A lot shall consist of lights manufactured under the same conditions during one production period.

4.4.3 *Rejection.* When one or more lights from a lot fail to conform to this specification, all components in the lot will be rejected.

4.5 **Test methods.**

4.5.1 *Examination of product.* Each light shall be examined to determine proper operation and conformance to all the requirements of this specification for which no tests are specified and to determine conformance to manufacturer's approved material, fabrication specifications, and drawings.

4.5.2 *Impact test.* The complete light with batteries shall be dropped from a height of 2 feet upon a hard, rigid surface, once on the bottom and once on each of the sides. Evidence of damage, except to the lamp, shall be cause for rejection.

4.5.3 *Low temperature.* The light shall be subjected to a temperature of 0° ±2°C for a period of 4 hours, and then checked for operation at this temperature for a period of 4 hours. Failure to operate satisfactorily shall be cause for rejection.

4.5.4 *Life.* With a new set of batteries installed, the light shall be operated continuously to determine its life. The active life of the light shall be considered as the period prior to the time it stops flashing or the cell voltage drops to 1.0 volt, whichever occurs first. If this time is less than that time specified in table I for the particular type of light being tested, the light shall be rejected.

4.5.5 *Photometric and chromaticity tests.* The light shall be subjected to photometric tests to determine conformance to all applicable photometric requirements of this specification, and to chromaticity tests to determine conformance to all applicable chromaticity requirements of Specification MIL–C–25050. These tests shall be made with the light

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

adjusted as received from the contractor and with the lamp operated at rated voltage (see 3.5.6.1 and 3.5.6.2). Failure to conform to the applicable requirements shall be cause for rejection.

4.5.6 *Rain.* The light shall be subjected to the Rain test, Procedure II of Specification MIL-E-5272. Failure to conform to the requirements specified in 3.5.4 shall be cause for rejection.

4.5.7 *Humidity.* The light shall be subjected to the humidity test, Procedure III of Specification MIL-E-5272. The light shall be visually inspected for evidence of corrosion. Failure to conform to the requirements specified in 3.5.5 shall be cause for rejection.

4.6 **Packaging, packing, and marking.** Preparation for delivery shall be inspected for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and packaging.** Preservation and packaging shall be level A or C as specified (see 6.3). Except as specified in 5.1.1.1, no preservation is required.

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Each marker light shall be unit packaged in accordance with Method III of Specification MIL-P-116 with a suitable wrap and placed within an interior box conforming to Specification PPP-B-676, PPP-B-665, or PPP-B-636.

5.1.1.2 *Intermediate packaging.* Intermediate packages shall contain the number of unit packages specified by the procuring activity. The intermediate packages shall then be placed in fiberboard boxes conforming to type I or II, class W5c or W6c of Specification PPP-B-636.

5.1.2 *Level C.*

5.1.2.1 Preservation and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 **Packing.** Packing shall be level A, B, or C as specified (see 6.3).

5.2.1 *Level A.* Marker lights packaged as specified in 5.1.1 shall be packed in overseas-type shipping containers conforming to Specification PPP-B-601, PPP-B-621, or MIL-B-10377. Boxes shall be strapped in accordance with the requirements of the appendix to the applicable box specification. The gross weight of the exterior shipping container when fully packed, shall not exceed approximately 200 pounds.

5.2.2 *Level B.* Marker lights packaged as specified in 5.1.2 shall be packed in domestic-type shipping containers conforming to Specification PPP-B-601, PPP-B-621, or MIL-B-10377. When specified by the procuring activity, intermediate packages may be shipped under this level without overpacking.

5.2.3 *Level C.* Marker lights packaged as specified in 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules, or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.** In addition to any special marking required by the contract or order, each unit package, intermediate package, and shipping container shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 **Intended use.** The lights covered by this specification are intended for use as emergency marker lights for applications requiring portability and short range visibility from the air, such as marking closed runways, helicopter operations, soft spots, or runway lighting on aircraft landing strips.

6.2 **Superseding data.** This specification supersedes Specifications MIL-L-7591B (ASG), dated 30 November 1959; MIL-L-52068(CE), dated 23 April 1959; MIL-L-

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.

19662(Aer), dated 1 November 1956; and MIL-L-19661(Aer), dated 15 October 1956.

6.3 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Quantity and type of lights required (see 1.2).

(c) Whether batteries should be furnished (see 3.5.2.2).

(d) Where the preproduction test samples should be sent, the activity responsible for testing, and instructions concerning the submittal of the test reports (see 4.3.1 and 4.3.3).

(e) Levels of packaging and packing and number of unit packages (see 5.1 and 5.2).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—TC
Navy—Wep

Preparing activity:
Navy—Wep

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:30Z
Check the source to verify that this is the current version before use.